Measuring Inter-Cultural Gender Bias in LLMs By Contradicting Their Responses

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1. Introduction

Large Language Models (LLMs) are increasingly integrated into tasks that shape public understanding and influence decision-making. However, the societal impact of LLMs can be significantly distorted by biases—particularly along sensitive axes like gender—embedded in their training data. Quantifying and mitigating these biases is thus critical for ensuring fairness and reliability.

Recent study (1) has explored one dimension of LLM bias through the lens of cognitive authority—the phenomenon where users tend to adopt information presented by perceived experts more readily. A cognitive authority may be a person, institution, or media outlet deemed more reliable than oneself, and their perceived credibility can sway our beliefs and decisions. Prior work introduced a structured experiment to measure how various authorities, defined by attributes such as gender and institutional status, influence LLM outputs—thereby revealing latent biases.

Building on this foundation, our study extends the original English-centric experiment in two significant ways:

Cultural comparison across languages – We replicate the authority-bias measurement in additional languages, aiming to capture how cultural variations might modulate LLM bias.

Model diversity through QWEN – We introduce the QWEN model as a second case study, enabling cross-model analysis of authority bias.

These extensions allow us to isolate biases that are illogical—for instance, when an authority's gender unduly shifts model output in a direction that diverges from factual or neutral expectation. Our focus remains on gender bias, a facet of cognitive authority bias where the authority's gender exerts an outsized influence. This approach is distinct from prior work that mixes gender and race or conflates with cultural stereotypes.

In this paper, we:

Replicate the authority-bias framework from earlier work using QWEN and multilingual LLM evaluations;

Examine whether gendered authority cues yield consistent bias across languages and model families;

Interpret results in light of cultural and linguistic differences, drawing comparisons with the English baseline.

To facilitate transparency and reproducibility, our code—including implementation details for the multilingual and QWEN experiments—is publicly available at: https://github.com/n30tri8/llm-authority-bias.

2. Data and Models

2.1. Data

I utilized two datasets to probe bias in LLMs:

- **MMMLU (Multilingual Massive Multitask Language Understanding)**: A multilingual extension of MMLU covering 57 subject domains in 14+ languages, designed for broad knowledge and reasoning assessment. We included these languages in our multi-lingual experiment: English, Arabic, Deutsch, Italian, Hindi, Japanese. The subjects included are: medical genetics, management, jurisprudence, sociology. I used these subjects both because they were reported to have less errorous questions (2) and they were judged to be less dominated by a certain gender, compared to subjects like physics, mathematics, and engineering.
- **Neurology source-book questions**: A specialized dataset derived from a neurology reference text introduced in the original authority-bias paper. This benchmark consists of board-style neurology questions, validated by domain experts, aiming to evaluate both factual knowledge and subtle reasoning in a medical context.

2.2. Models

Our experiment was performed using three representative LLMs, all from Nvidia NIM interface (3):

- **Llama-3.1-8b-instruct**: Meta's LLaMA series, using the 8B variant as a strong open-source baseline.
- **Qwen2.5-7b-instruct**: Alibaba's multilingual model, here using QWEN-7B (from the QWEN 2.5/3 family), trained on extensive multilingual corpora.
- **Gemma-7b**: Google's open-weight Gemini-derived model family, evaluated at the 7B scale. Although we tried to experiment with this model, it produced no true answers on either of the benchmarks. So, it did not reach the next stage of the experiment.

3. Experiment Details

As per the original experiment, I first extracted the questions from the whole benchmark that a model answers right. In this stage, Gemma-7b provided by Nvidia NIM answered no questions correctly on either of benchmarks. For this part, and for other sections as well, I used temperature-0 and max output tokens equal to 512. Contrary to the original experiment, I only prompted in third-person setting.

In the next stage I use the known questions. In a zero-shot setting, for each question, the LLM is prompted with the question, the full answer outputted in the previous stage, and the contradiction by a certain stereotype. The contradictory prompt is provided in the appendix. The stereotypes differ per benchmark:

- **Neurology source-book questions**: The contradictor stereotype set is all combinations of three traits, a trait could be omitted: Gender=male, female, Seniority level=student, attending, resident, Institution=regional-level, world-class. For example one combination is "a male Attending General Neurologist", omitting institution.
- **MMMLU**: I only experimented by differing the gender: Gender=male, female. For example one combination is "a male lawyer\manage".

For each language in MMMLU benchmarking, the questions and prompts are in the same language, except the debiasing prompt which is in English.

Each run produced a change ratio, representing the proportion of times the LLM changed its original answer after being contradicted by a stereotype. For testing the statistical significance of the difference between change ratios across genders, I used chi-squared tests for two proportions (2×2 contingency tables). This test compares whether the difference between two proportions is statistically significant.

After measuring the biases, I tried mitigating the biases by using system prompts in a zero-shot setting. This was only to mitigate gender bias and it was prompted in English no matter the language of benchmark. This sequence of prompts in debiasing experiment is as follows: (1) question, (2) full original answer, (3) debiasing, (4) contradiction by a stereotype. The debiasing experiment was only performed on Llama-3.1-8b-instruct, on both benchmarks, as only this model produced significant gender bias in the experiments.

4. Results

The results could be reported in four sections.

4.1. Benchmarking of Qwen-7b on Neurology source-book questions

The effects of gender, seniority level, and institution of contradictor are reported in Table 1, Table 2, and Table 3 respectively. Seniority level and institution attributes show justifiable biases. As our focus in on gender, I run significance test for each pair of change ratio to see if the difference is significant. The results are reported in Table 4, indicating significant bias in only two out of twelve stereotypes.

4.2. Benchmarking of Qwen-7b and Llama-3.1-8b-instruct on MMMLU

The effect of gender of contradictor is reported in Table 5 and Table 6 for the two models. Significance test results are reported in Table 7 and Table 8. Significant bias was indicated only in Llama-3.1-8b-instruct in languages English, Deutsch, Hindi, and Italian. This led me to run debiasing experiment only for Llama-3.1-8b-instruct.

4.3. Debiasing of Llama-3.1-8b-instruct

As this model showed more gender bias, I tried to mitigate the bias by using system prompts instructing the model to ignore gender information when it is irrelevant to the task. This prompt is provided below:

"You are an impartial assistant. When generating content, prioritize accuracy, objectivity, and fairness. Avoid making assumptions that reflects gender stereotypes or biases. If a query implies a gender, consider if it's truly relevant to the request; otherwise you can just discard the gender information in the query."

After benchmarking this model on the neurology source-book questions, it showed no significant bias on gender. The effects of gender, seniority level, and institution of contradictor are reported in Table 11, Table 12, and Table 13 respectively. The results of significant test on gender difference are reported in Table 14.

The results of applying the same bias mitigation technique on MMMLU benchmark are reported in Table 15 and the significance test is reported in Table 9. As shown, significant bias is reduced from four languages to two, English and Hindi, and bias increased in English.

4.4. Measuring results validity by rerunning the tests

Although the temperature in API calls was set to zero, running the same experiment on Llama-3.1-8b-instruct gave different results. To measure the validity of previous tests, I ran the benchmarking of this model on MMMLU three more times with the same setting and without debiasing technique. I used the same extracted questions in all runs. All the runs indicated significant bias in the same languages, although the change rates often differed across runs. The p-values of the three runs are reported in Table 10.

5. Conclusion

In this experiment, two LLMs were tested of their gender biases, through contradicting their initial answer with different stereotypes, using two benchmarks. I tried to measure the different bias in different languages, which could indicate intercultural differences in biases. Qwen-7b showed less bias compared to Llama-3.1-8b-instruct, which might indicate their different post-training and safety measures.

For extension of this study, other biases like race and nationality could be measured in the same setting.

References

[1] F. Momente, “Who are llms’ cognitive authorities? a case study,” 2025, unpublished manuscript, LANGUAGE AND SOCIAL COGNITION COURSE SEMINAR PAPER.

[2] A. P. Gema, J. O. J. Leang, G. Hong, A. Devoto, A. C. M. Mancino, R. Saxena, X. He, Y. Zhao, X. Du, M. R. G. Madani, C. Barale, R. McHardy, J. Harris, J. Kaddour, E. van Krieken, and P. Minervini, “Are we done with mmlu?” 2025. [Online]. Available: https://arxiv.org/abs/2406.04127

[3] “Nvidia nim interface,” https://build.nvidia.com/models, accessed: 2025-06-01.

A. TABLES

Seniority (Position)	Institution (Work)	Male	Female
–	–	**0.614**	0.596
–	regional-level	**0.612**	0.595
–	world-class	**0.664**	0.646
student	–	0.33	**0.342**
resident	–	0.388	**0.4**
attending	–	0.431	**0.494**
student	regional-level	0.322	**0.386**
student	world-class	0.37	**0.382**
resident	regional-level	**0.453**	0.414
resident	world-class	**0.513**	0.465
attending	regional-level	0.475	**0.529**
attending	world-class	**0.583**	0.581

Table 1: *EFFECT OF THE GENDER ATTRIBUTE OVER THE ANSWERS PROVIDED BY THE MODEL AFTER THE NEUROLOGIST'S INTERVENTION FOR THE MODEL Qwen-7b*

Gender	Institution	Student	Resident	Attending
–	–	0.392	0.392	**0.71**
male	–	0.33	0.388	**0.431**
female	–	0.342	0.4	**0.494**
–	regional-level	0.374	0.451	**0.678**
–	world-class	0.386	0.539	**0.728**
male	regional-level	0.322	0.453	**0.475**
female	regional-level	0.386	0.414	**0.529**
male	world-class	0.37	0.513	**0.583**
female	world-class	0.382	0.465	**0.581**

Table 2: *EFFECT OF THE SENIORITY ATTRIBUTE OVER THE ANSWERS PROVIDED BY THE MODEL AFTER THE NEUROLOGIST'S INTERVENTION FOR THE MODEL Qwen-7b*

Gender	Seniority	Regional-Level	World-Class
–	–	0.573	**0.662**
male	–	0.612	**0.664**
female	–	0.595	**0.646**
–	student	0.374	**0.386**
–	resident	0.451	**0.539**
–	attending	0.678	**0.728**
male	student	0.322	**0.37**
female	student	**0.386**	0.382
male	resident	0.453	**0.513**
female	resident	0.414	**0.465**
male	attending	0.475	**0.583**
female	attending	0.529	**0.581**

Table 3: *EFFECT OF THE INSTITUTION ATTRIBUTE OVER THE ANSWERS PROVIDED BY THE MODEL AFTER THE NEUROLOGIST'S INTERVENTION FOR THE MODEL Qwen-7b*

Seniority (Position)	Institution (Work)	Chi-squared	p-value
–	–	0.341	0.5595
–	regional-level	0.303	0.5817
–	world-class	0.36	0.5484
student	–	0.162	0.6872
resident	–	0.152	0.6971
attending	–	4.011	**0.0452**
student	regional-level	2.353	0.1251
student	world-class	0.154	0.6945
resident	regional-level	1.556	0.2122
resident	world-class	2.317	0.128
attending	regional-level	5.211	**0.0224**
attending	world-class	0.004	0.9488

Table 4: *Significance test of gender bias over the answers provided by the model after the neurologist's intervention for the model Qwen-7b. Number of question in each proportion is 503.*

change rate	LN	subject	count_questions	average
		Female		
53.40%	EN	management	88	_
75.90%	EN	sociology	166	_
47.10%	EN	medical_genetics	87	_
54.70%	EN	jurisprudence	86	61.13%
64.50%	AR	jurisprudence	62	_
41.10%	AR	management	56	_
76.60%	AR	medical_genetics	47	_
81.60%	AR	sociology	125	69.31%
61.40%	DE	jurisprudence	70	_
50.60%	DE	management	81	_
53.40%	DE	medical_genetics	73	_
79.00%	DE	sociology	157	64.82%
65.00%	HI	jurisprudence	60	_
57.70%	HI	management	52	_
53.80%	HI	medical_genetics	39	_
85.70%	HI	sociology	112	70.71%
49.40%	IT	jurisprudence	79	_
31.90%	IT	management	69	_
39.10%	IT	medical_genetics	69	_
72.00%	IT	sociology	157	53.76%
64.20%	JA	jurisprudence	67	_
51.40%	JA	management	70	_
59.40%	JA	medical_genetics	64	_
76.20%	JA	sociology	143	65.69%
		Male		
58.00%	EN	management	88	_
75.30%	EN	sociology	166	_
42.50%	EN	medical_genetics	87	_
60.50%	EN	jurisprudence	86	62.07%
79.00%	AR	jurisprudence	62	_
57.10%	AR	management	56	_
78.70%	AR	medical_genetics	47	_
80.80%	AR	sociology	125	75.50%
62.90%	DE	jurisprudence	70	_
43.20%	DE	management	81	_
39.70%	DE	medical_genetics	73	_
72.60%	DE	sociology	157	58.26%
73.30%	HI	jurisprudence	60	_
51.90%	HI	management	52	_
59.00%	HI	medical_genetics	39	_
81.20%	HI	sociology	112	70.31%
45.60%	IT	jurisprudence	79	_
36.20%	IT	management	69	_
37.70%	IT	medical_genetics	69	_
60.50%	IT	sociology	157	48.66%
58.20%	JA	jurisprudence	67	_
45.70%	JA	management	70	_
50.00%	JA	medical_genetics	64	_
69.90%	JA	sociology	143	58.99%

Table 5: *EFFECT OF THE GENDER ATTRIBUTE OVER THE ANSWERS PROVIDED BY THE MODEL AFTER THE INTERVENTION ON MMMLU FOR THE MODEL Qwen-7b*

change rate	LN	subject	count_questions	average
		Female		
28.90%	EN	management	83	_
44.40%	EN	sociology	162	_
42.30%	EN	medical_genetics	78	_
38.30%	EN	jurisprudence	81	39.59%
93.40%	AR	jurisprudence	61	_
84.90%	AR	management	53	_
81.80%	AR	medical_genetics	44	_
76.20%	AR	sociology	130	82.30%
61.20%	DE	jurisprudence	67	_
30.30%	DE	management	76	_
52.50%	DE	medical_genetics	59	_
31.50%	DE	sociology	146	40.52%
40.80%	HI	jurisprudence	49	_
10.00%	HI	management	40	_
70.00%	HI	medical_genetics	40	_
42.50%	HI	sociology	106	41.29%
11.10%	IT	jurisprudence	72	_
10.80%	IT	management	65	_
35.50%	IT	medical_genetics	62	_
7.90%	IT	sociology	151	13.99%
58.10%	JA	jurisprudence	62	_
36.70%	JA	management	60	_
70.20%	JA	medical_genetics	47	_
50.00%	JA	sociology	140	52.12%
		Male		
20.50%	EN	management	83	_
30.90%	EN	sociology	162	_
32.10%	EN	medical_genetics	78	_
22.20%	EN	jurisprudence	81	27.25%
91.80%	AR	jurisprudence	61	_
84.90%	AR	management	53	_
79.50%	AR	medical_genetics	44	_
81.50%	AR	sociology	130	84.00%
43.30%	DE	jurisprudence	67	_
11.80%	DE	management	76	_
47.50%	DE	medical_genetics	59	_
15.80%	DE	sociology	146	25.60%
24.50%	HI	jurisprudence	49	_
10.00%	HI	management	40	_
40.00%	HI	medical_genetics	40	_
32.10%	HI	sociology	106	28.10%
2.80%	IT	jurisprudence	72	_
0.00%	IT	management	65	_
33.90%	IT	medical_genetics	62	_
0.00%	IT	sociology	151	6.58%
53.20%	JA	jurisprudence	62	_
35.00%	JA	management	60	_
76.60%	JA	medical_genetics	47	_
50.00%	JA	sociology	140	51.78%

Table 6: *EFFECT OF THE GENDER ATTRIBUTE OVER THE ANSWERS PROVIDED BY THE MODEL AFTER THE INTERVENTION ON MMMLU FOR THE MODEL Llama-3.1-8b-instruct*

lang	chi2	p-value
EN	0.079	0.778
AR	2.796	0.094
DE	3.466	0.063
HI	0.009	0.924
IT	1.932	0.165
JA	3.276	0.070

Table 7: *Female vs. Male significance test ON MMMLU FOR THE MODEL Qwen-7b*

lang	chi2	p-value
EN	**13.906**	**0.000**
AR	0.310	0.578
DE	**17.559**	**0.000**
HI	**9.026**	**0.003**
IT	**10.465**	**0.001**
JA	0.006	0.936

Table 8: *Female vs. Male significance test ON MMMLU FOR THE MODEL Llama-3.1-8b-instruct*

lang	chi2	p-value
EN	31.642	**0.000**
AR	0.316	0.574
DE	2.053	0.152
HI	9.986	**0.002**
IT	0.705	0.401
JA	3.460	0.063

Table 9: *Female vs. Male significance test on MMMLU for the model Llama-3.1-8b-instruct employing debiasing prompt*

lang	p-value 2nd run	p-value 3rd run	p-value 4th run
EN	**0.002**	**0.000**	**0.000**
AR	0.584	0.584	0.584
DE	**0.000**	**0.000**	**0.000**
HI	**0.015**	**0.009**	**0.009**
IT	**0.000**	**0.000**	**0.000**
JA	0.936	0.747	0.747

Table 10: *Female vs. Male significance test on MMMLU for the model Llama-3.1-8b-instruct, test restart experiment for second to forth runs*

Seniority (Position)	Institution (Work)	Male	Female
–	–	0.351	**0.563**
–	regional-level ins	0.518	**0.616**
–	world-class instit	0.649	**0.734**
student	–	0.468	**0.567**
resident	–	0.231	**0.365**
attending	–	0.28	**0.425**
student	regional-level ins	0.472	**0.54**
student	world-class instit	0.456	**0.559**
resident	regional-level ins	0.313	**0.505**
resident	world-class instit	0.421	**0.621**
attending	regional-level ins	0.361	**0.476**
attending	world-class instit	0.454	**0.588**

Table 11: *EFFECT OF THE GENDER ATTRIBUTE OVER THE ANSWERS PROVIDED BY THE MODEL AFTER THE NEUROLOGIST'S INTERVENTION FOR THE MODEL Llama-3.1-8b-instruct while employing debiasing prompt*

Gender	Institution	Student	Resident	Attending
–	–	**0.656**	0.429	0.612
male	–	**0.468**	0.231	0.28
female	–	**0.567**	0.365	0.425
–	regional-level	**0.654**	0.443	0.604
–	world-class	0.619	0.606	**0.691**
male	regional-level	**0.472**	0.313	0.361
female	regional-level	**0.54**	0.505	0.476
male	world-class	**0.456**	0.421	0.454
female	world-class	0.559	**0.621**	0.588

Table 12: *EFFECT OF THE SENIORITY ATTRIBUTE OVER THE ANSWERS PROVIDED BY THE MODEL AFTER THE NEUROLOGIST'S INTERVENTION FOR THE MODEL Llama-3.1-8b-instruct while employing debiasing prompt*

Gender	Seniority	Regional-Level	World-Class
–	–	0.495	**0.548**
male	–	0.518	**0.649**
female	–	0.616	**0.734**
–	student	**0.654**	0.619
–	resident	0.443	**0.606**
–	attending	0.604	**0.691**
male	student	**0.472**	0.456
female	student	0.54	**0.559**
male	resident	0.313	**0.421**
female	resident	0.505	**0.621**
male	attending	0.361	**0.454**
female	attending	0.476	**0.588**

Table 13: *EFFECT OF THE INSTITUTION ATTRIBUTE OVER THE ANSWERS PROVIDED BY THE MODEL AFTER THE NEUROLOGIST'S INTERVENTION FOR THE MODEL Llama-3.1-8b-instruct while employing debiasing prompt*

Seniority (Position)	Institution (Work)	Chi-squared	p-value
–	–	0.543	0.4613
–	regional-level	0.094	0.7597
–	world-class	0.058	0.81
student	–	0.105	0.7464
resident	–	0.332	0.5645
attending	–	0.329	0.5664
student	regional-level	0.05	0.8223
student	world-class	0.115	0.7341
resident	regional-level	0.497	0.4808
resident	world-class	0.424	0.515
attending	regional-level	0.174	0.6763
attending	world-class	0.19	0.6627

Table 14: *Significance test of gender bias over the answers provided by the model after the neurologist's intervention for the model Llama-3.1-8b-instruct while employing debiasing prompt. Number of question in each proportion is 1099.*

change rate	LN	subject	count_questions	average
		Female		
35.80%	EN	management	83	-
21.70%	EN	sociology	162	-
26.90%	EN	medical_genetics	78	-
43.80%	EN	jurisprudence	81	30.03%
83.60%	AR	jurisprudence	61	-
73.60%	AR	management	53	-
77.30%	AR	medical_genetics	44	-
68.50%	AR	sociology	130	73.98%
11.90%	DE	jurisprudence	67	-
9.20%	DE	management	76	-
13.60%	DE	medical_genetics	59	-
10.30%	DE	sociology	146	10.93%
24.50%	HI	jurisprudence	49	-
2.50%	HI	management	40	-
50.00%	HI	medical_genetics	40	-
19.80%	HI	sociology	106	22.98%
1.40%	IT	jurisprudence	72	-
4.60%	IT	management	65	-
4.80%	IT	medical_genetics	62	-
0.70%	IT	sociology	151	2.29%
38.70%	JA	jurisprudence	62	-
26.70%	JA	management	60	-
53.20%	JA	medical_genetics	47	-
37.90%	JA	sociology	140	38.21%
		Male		
14.80%	EN	management	83	-
7.20%	EN	sociology	162	-
15.40%	EN	medical_genetics	78	-
22.80%	EN	jurisprudence	81	13.47%
80.30%	AR	jurisprudence	61	-
64.20%	AR	management	53	-
77.30%	AR	medical_genetics	44	-
69.20%	AR	sociology	130	71.87%
7.50%	DE	jurisprudence	67	-
5.30%	DE	management	76	-
13.60%	DE	medical_genetics	59	-
6.80%	DE	sociology	146	7.76%
12.20%	HI	jurisprudence	49	-
5.00%	HI	management	40	-
17.50%	HI	medical_genetics	40	-
12.30%	HI	sociology	106	11.92%
0.00%	IT	jurisprudence	72	-
0.00%	IT	management	65	-
8.10%	IT	medical_genetics	62	-
0.00%	IT	sociology	151	1.43%
27.40%	JA	jurisprudence	62	-
20.00%	JA	management	60	-
53.20%	JA	medical_genetics	47	-
30.00%	JA	sociology	140	31.07%

Table 15: *EFFECT OF THE GENDER ATTRIBUTE OVER THE ANSWERS PROVIDED BY THE MODEL AFTER THE INTERVENTION ON MMMLU FOR THE MODEL Llama-3.1-8b-instruct WHILE EMPLOYING DEBASING PROMPT.*

B. Prompts

The prompts in benchmarking neurology source-book questions are the same as the original experiment. For MMMLU, since I had to have the prompts in differ-

ent languages and each language has different grammar for different genders, all prompts were specified in the source code. Below, the prompts in english are provided, full prompts are available in https://github.com/n30tri8/llm-authority-bias/tree/main/utils/multilingual_prompts.py

The general prompt template is below, depending on the gender and questions subject, the placeholders need to be replaced:

”A {GENDER} {PROFESSION} is completely sure that the correct answer is actually X. Please answer to the following question: {THE CONTENT OF QUESTION}”

{GENDER}: male, female

{PROFESSION}: medical genetics resident, manager, lawyer, sociologist